CONTROL ENGINEERING

1. a) Cayley-Hamilton Theorem (extensively discussed in the lectures) states that

$$p(A) = 0,$$

where

$$p(\lambda) = \det(\lambda I - A) = \lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_1\lambda + \alpha_0.$$

As a result

$$A^n + \alpha_{n-1}A^{n-1} + \cdots + \alpha_1 A + \alpha_0 I.$$

Multiplying by A^{-1}, which is well defined, yields

$$A^{n-1} + \alpha_{n-1}A^{n-2} + \cdots + \alpha_1 I + \alpha_0 A^{-1}.$$

From this, since $\alpha_0 \neq 0$, one derives an expression for A^{-1} in terms of the coefficients α_i and the powers of A. [4 marks]

b) A direct integration of the differential equations yields

$$x_1(1) = \frac{1}{2}\alpha, \qquad x_2(1) = \alpha,$$

and

$$x_1(2) = \frac{3}{2}\alpha + \frac{1}{2}\beta, \qquad x_2(2) = \alpha + \beta.$$

Hence, setting $\alpha = 4$ and $\beta = -4$, yields the desired final condition. [6 marks]

Note that the point $=[4 \;\; 0]^\top$ is an equilibrium of the system for $u = 0$. Hence, the selection $u(t) = 0$, for all $t \geq 2$, guarantees that $x(t) = x(2)$ for all $t \geq 2$. [2 marks]

c) The equilibrium points are obtained solving the equations

$$0 = -x_1 + x_1 x_2, \qquad 0 = -x_2 - x_1 x_2 + 2,$$

which yields the points $\bar{x} = [0 \;\; 2]^\top$ and $\tilde{x} = [1 \;\; 1]^\top$. [4 marks]

The matrices describing the linearized systems around $\tilde{x}$ are

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 2 & 2 \end{bmatrix}.$$

[6 marks]

The characteristic polynomial of A is $p(\lambda) = \lambda^2 + 2\lambda + 1$, which has all roots with negative real parts. As a result, the $\tilde{x}$ equilibrium is a locally asymptotically stable equilibrium of the nonlinear system. [4 marks]

Answers

d) The matrices A, B, and C of the system are

$$A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & -2 \end{bmatrix}.$$

Note that $A^2 = 0$, which implies controllability of the system. [2 marks]

The observability matrix of the system is

$$\mathscr{O} = \begin{bmatrix} 1 & -2 \\ 3 & 3 \end{bmatrix},$$

which is full rank, hence the system is observable. [2 marks]

The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} \alpha & \alpha+\beta \\ \beta & -\alpha-\beta \end{bmatrix}.$$

Note that $\det \mathscr{R} = -(\alpha+\beta)^2$, hence the system is reachable provided $\alpha \neq -\beta$. [2 marks]

e) Note that (with obvious notation)

$$\begin{aligned} e^+ &= x^+ - w^+, \\ &= x + u - (z^+ + cx), \\ &= x + u - az - bu - cx, \\ &= (1-c)(x+w) + (1-b)u - az. \end{aligned}$$

Setting $a = 0$, $b = 1$, and $c = 1$, yields $e^+ = 0$. As a result, for any initial conditions $e(1) = 0$, hence $K = 1$. [8 marks]

Answers

2. a) Note that

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2k & 0 & k & 0 \\ 0 & 0 & 0 & 1 \\ k & 0 & -2k & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

[8 marks]

b) The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} 0 & 0 & 1 & 0 & \cdots \\ 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \end{bmatrix},$$

which is (trivially) full rank (note that there is no need to compute A^2B and A^3B), hence the system is controllable. [6 marks]

c) In the new coordinates the system is described by the equations

$$\dot{z}_1 = z_2, \qquad \dot{z}_2 = -3kz_1 + u_1 - u_2, \qquad \dot{z}_3 = z_4, \qquad \dot{z}_4 = kz_1 - kz_3 + u_2,$$

hence

$$\tilde{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -3k & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ k & 0 & -k & 0 \end{bmatrix}, \qquad \tilde{B} = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

[6 marks]

The characteristic polynomial of $\tilde{A}$, which is a block lower triangular matrix is $p(\lambda) = (\lambda^2 + k)(\lambda^2 + 3k)$. Note that since $k > 0$ the matrix $\tilde{A}$, hence A, has four eigenvalues on the imaginary axis, all with multiplicity one. The system is therefore stable. [4 marks]

d) Setting $u_1 = u_2 = u$ yields

$$\dot{z}_1 = z_2, \qquad \dot{z}_2 = -3kz_1, \qquad \dot{z}_3 = z_4, \qquad \dot{z}_4 = kz_1 - kz_3 + u.$$

Hence, the system is already decomposed into reachable (the $(z_3,\ z_4)$ subsystem) and unreachable (the $(z_1,\ z_2)$ subsystem) subsystems. The unreachable modes are therefore $\pm j\sqrt{3k}$, hence the system is not stabilizable. [6 marks]

Answers

3. a) Note that

$$A = \begin{bmatrix} -2 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

i) The reachability matrix of the system is

$$\mathscr{R} = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix},$$

hence the system is reachable (and controllable), that is it is possible to reach from $x = 0$ any $x(T)$ in any given time $T > 0$. [4 marks]

ii) Setting u_1 constant and solving the equations $\dot{x}_1 = \dot{x}_2 = \dot{x}_3 = 0$ yields the equilibrium point $\bar{x} = \begin{bmatrix} \frac{3}{4} u_1 & \frac{1}{2} u_1 & \frac{1}{4} u_1 \end{bmatrix}^\top$. [4 marks]

iii) Reachability implies that any state x_f can be reached in some given time $T > 0$, but not the state of the system can remain in x_f for all $t \geq T$. If x_f is in addition an equilibrium point, it can be reached in any $T > 0$ and then u_1 can be selected such that $x(t) = x_f$ for all $t \geq T$. [4 marks]

b) The observability matrix of the system is

$$\mathscr{O} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -2 & 1 \\ -4 & 6 & -4 \end{bmatrix}.$$

Note that the first and third column are the same, and the first and second column are linearly independent. The system is therefore not observable, and it has one unobservable mode. To determine the unobservable mode consider the observability pencil

$$\begin{bmatrix} 0 & 1 & 0 \\ \lambda+2 & -1 & 0 \\ -1 & \lambda+2 & -1 \\ 0 & -1 & \lambda+2 \end{bmatrix}.$$

By eliminating the first row and the second column, one sees that the observability pencil loses rank for $\lambda = -2$, which is therefore the unobservable mode. [8 marks]

To design an observer consider the matrix $A_{oi} = A - LC$, that is

$$A_{oi} = \begin{bmatrix} -2 & 1-L_1 & 0 \\ 1 & -2-L_2 & 1 \\ 0 & 1-L_3 & -2 \end{bmatrix}.$$

Setting $L_2 = 0$ and $L_1 = 2 - L_3$ yields the desired eigenvalues. Note that it is possible to assign the eigenvalues of the matrix A_{oi} to the desired values, even if the system is not observable, because the unobservable mode coincides with one of the desired eigenvalues of the estimation error system. [8 marks]

c) The output injection matrix L achieving the desired estimation error dynamics is not unique since L has, *de facto*, to assign only two eigenvalues. The third one is fixed to -2. [2 marks]

Answers